Conic Sections General: Questions and Shortened Solutions

Question 01

English

Let C be an ellipse having a focus at the origin O and the corresponding directrix parallel to the y-axis. If C passes through the points $A(4,3)$ and $B(0,-4)$, then the x-coordinate of the vertex of C nearer to this directrix is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C एक दीर्घवृत्त है जिसकी एक नाभि मूलबिन्दु O पर है तथा संगत नियता y-अक्ष के समांतर है। यदि C बिन्दुओं $A(4,3)$ तथा $B(0,-4)$ से होकर जाता है, तो C के उस शीर्ष का x-निर्देशांक, जो इस नियता के अधिक निकट है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let the directrix be $x = d$ $(d \neq 0)$ and the eccentricity e. Focus-directrix property $PS = e \cdot PM$:

At A: $5 = e|4 - d|$; at B: $4 = e|d| \Rightarrow 5|d| = 4|4 - d|$

$5d = 4(4 - d) \Rightarrow d = \dfrac{16}{9},\ e = \dfrac{9}{4} > 1$ (hyperbola, rejected); $5d = -4(4 - d) \Rightarrow d = -16,\ e = \dfrac{1}{4} < 1$ (ellipse)

$C:\ x^2 + y^2 = \dfrac{1}{16}(x + 16)^2 \Rightarrow 15x^2 - 32x + 16y^2 = 256$

The axis is the x-axis (through the focus, perpendicular to the directrix): $y = 0 \Rightarrow 15x^2 - 32x - 256 = 0 \Rightarrow x = \dfrac{16}{3}$ or $x = -\dfrac{16}{5}$

The vertex nearer to $x = -16$ is $\left(-\dfrac{16}{5}, 0\right)$

Final Answer: -3.20

Question 02

English

For integers p and q with $1 \leq p \leq 20$ and $1 \leq q \leq 20$, let

$$E_{p,q} = \{z \in \mathbb{C} : |z - p| + |z - qi| = 20\}, \qquad H_{p,q} = \{z \in \mathbb{C} : \big||z - p| - |z - qi|\big| = 10\},$$

where $i = \sqrt{-1}$. Let N be the number of ordered pairs (p, q) for which $E_{p,q}$ is an ellipse and $H_{p,q}$ is a hyperbola. Then the value of N is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

पूर्णांकों p तथा q के लिए, जहाँ $1 \leq p \leq 20$ तथा $1 \leq q \leq 20$ हैं, माना कि

$$E_{p,q} = \{z \in \mathbb{C} : |z - p| + |z - qi| = 20\}, \qquad H_{p,q} = \{z \in \mathbb{C} : \big||z - p| - |z - qi|\big| = 10\},$$

जहाँ $i = \sqrt{-1}$ है। माना कि N उन क्रमित युग्मों (p, q) की संख्या है जिनके लिए $E_{p,q}$ एक दीर्घवृत्त है तथा $H_{p,q}$ एक अतिपरवलय है। तब N का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both loci have foci p and qi, with focal distance $2c = |p - qi| = \sqrt{p^2 + q^2}$

$E_{p,q}$: $2a = 20$, $e = \dfrac{\sqrt{p^2+q^2}}{20}$. Ellipse iff $e < 1$; if $p^2 + q^2 = 400$ it is the segment joining the foci (degenerate), if greater it is empty

$H_{p,q}$: $2a = 10$, $e = \dfrac{\sqrt{p^2+q^2}}{10}$. Hyperbola iff $e > 1$; if $p^2 + q^2 = 100$ it is two rays (degenerate), if smaller it is empty (triangle inequality)

So we need $100 < p^2 + q^2 < 400$ with $1 \le p, q \le 20$

Pairs with $p^2+q^2 < 400$ (so $p \le 19$): for $p = 1, \ldots, 19$ the counts of q are $19, 19, 19, 19, 19, 19, 18, 18, 17, 17, 16, 15, 15$ total 292 (the boundary pairs $(12, 16), (16, 12)$ are left out)

Pairs with $p^2 + q^2 \le 100$: for $p = 1, \ldots, 9$ the counts are $9, 9, 9, 9, 8, 8, 7, 6, 4$, total 69 (including the boundary pairs $(6, 8), (8, 6)$)

$N = 292 - 69 = 223$

Final Answer: 223

Question 03

English

Let E be the ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ and, for a non-zero real number k, let H_k be the hyperbola $xy = k$. Let S be the set of all non-zero real numbers k for which E and H_k have exactly two common tangents. Then the product of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ है तथा किसी शून्येतर वास्तविक संख्या k के लिए H_k अतिपरवलय $xy = k$ है। माना कि S उन सभी शून्येतर वास्तविक संख्याओं k का समुच्चय है जिनके लिए E तथा H_k की ठीक दो उभयनिष्ठ स्पर्श रेखाएँ हैं। तब S के सभी अवयवों का गुणनफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Vertical and horizontal lines are never tangent to $xy = k$, so a common tangent is $y = mx + c$, $m \ne 0$

Tangent to E: $c^2 = 9m^2 + 4$. Tangent to H_k: $mx^2 + cx - k = 0$ has equal roots $\Rightarrow c^2 = -4mk$

$\Rightarrow 9m^2 + 4km + 4 = 0$; each real root m gives two lines $c = \pm\sqrt{9m^2 + 4}$

Discriminant $16k^2 - 144$: $|k| > 3$ gives two slopes (four tangents), $|k| < 3$ none, $|k| = 3$ a double root $m = -\dfrac{2k}{9}$ (exactly two tangents)

$S = \{3, -3\}$, product $= -9$

Final Answer: -9

Question 04

English

Let P be the parabola $y^2 = 4x$ and C the circle $x^2 + y^2 - 2x - 4 = 0$. The common tangents of P and C meet at the point T, and they touch C at the points B_1 and B_2. If K is the centre of C, then the area of the quadrilateral TB_1KB_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P परवलय $y^2 = 4x$ है तथा C वृत्त $x^2 + y^2 - 2x - 4 = 0$ है। P तथा C की उभयनिष्ठ स्पर्श रेखाएँ बिन्दु T पर मिलती हैं तथा C को बिन्दुओं B_1 तथा B_2 पर स्पर्श करती हैं। यदि K, C का केन्द्र है, तो चतुर्भुज TB_1KB_2 का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$C : (x-1)^2 + y^2 = 5$, centre $K(1,0)$ (the focus of P), radius $\sqrt{5}$

Tangent to P: $y = mx + \dfrac{1}{m}$; distance from $K = \dfrac{|m + 1/m|}{\sqrt{1+m^2}} = \dfrac{\sqrt{1+m^2}}{|m|} = \sqrt{5} \Rightarrow m^2 = \dfrac{1}{4}$ (the tangent $x = 0$ is at distance 1, not a common tangent)

Common tangents $x - 2y + 4 = 0$ and $x + 2y + 4 = 0$, meeting at $T(-4, 0)$

Normal to C at B_i passes through K, so $KB_i \perp TB_i$: $B_1, B_2 = (0, \pm 2)$ (feet of the perpendiculars from K), and $TB_i = \sqrt{TK^2 - 5} = \sqrt{25 - 5} = 2\sqrt{5}$

Area $= 2 \cdot \dfrac{1}{2} \cdot KB_1 \cdot TB_1 = \sqrt{5} \cdot 2\sqrt{5} = 10$

Final Answer: 10

Question 05

English

Let P_1 be the parabola $y^2 = 4(x+1)$. Let P_2 be the parabola that has the same focus and the same axis as P_1 and opens in the opposite direction. If the length of the common chord of P_1 and P_2 is 6, then the distance between the vertices of P_1 and P_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P_1 परवलय $y^2 = 4(x+1)$ है। माना कि P_2 वह परवलय है जिसकी नाभि तथा अक्ष P_1 के समान हैं तथा जो विपरीत दिशा में खुलता है। यदि P_1 तथा P_2 की उभयनिष्ठ जीवा की लम्बाई 6 है, तो P_1 तथा P_2 के शीर्षों के बीच की दूरी _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

P_1: vertex $(-1, 0)$, $a = 1$, focus $(0, 0)$, axis the x-axis

P_2 opens towards $-x$ with focus $(0, 0)$: vertex $(b, 0)$, $b > 0$, so $P_2 : y^2 = -4b(x - b)$

Common chord (subtracting): $4(x+1) = -4b(x-b) \Rightarrow (1+b)x = b^2 - 1 \Rightarrow x = b - 1$

Then $y^2 = 4b$, so the chord length is $4\sqrt{b} = 6 \Rightarrow b = \dfrac{9}{4}$

Distance between vertices $= 1 + \dfrac{9}{4} = \dfrac{13}{4}$

Final Answer: 3.25

Question 06

English

Consider the ellipse $E : \dfrac{x^2}{4} + y^2 = 1$. Let C_1 and C_2 be the circles with centres $\left(-\dfrac{3}{10}, 0\right)$ and $\left(\dfrac{6}{5}, 0\right)$ respectively, each of which touches E at two distinct points. Then the length of the common chord of C_1 and C_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दीर्घवृत्त $E : \dfrac{x^2}{4} + y^2 = 1$ पर विचार कीजिए। माना कि C_1 तथा C_2 क्रमशः केन्द्रों $\left(-\dfrac{3}{10}, 0\right)$ तथा $\left(\dfrac{6}{5}, 0\right)$ वाले वे वृत्त हैं जिनमें से प्रत्येक E को दो भिन्न बिन्दुओं पर स्पर्श करता है। तब C_1 तथा C_2 की उभयनिष्ठ जीवा की लम्बाई _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At a point of contact the radius is a common normal of the circle and E, so the centre lies on the normal of E

Normal at $(2\cos\theta, \sin\theta)$: $2x\sec\theta - y\csc\theta = 3$ meets $y = 0$ at $x = \dfrac{3}{2}\cos\theta = \dfrac{3}{4}x_0$ $(e^2 = \dfrac{3}{4})$, so $x_0 = \dfrac{4}{3}h$

C_1: $h = -\dfrac{3}{10}$, $x_0 = -\dfrac{2}{5}$, $y_0^2 = 1 - \dfrac{1}{25} = \dfrac{24}{25}$, $r_1^2 = \left(\dfrac{1}{10}\right)^2 + \dfrac{24}{25} = \dfrac{97}{100}$

C_2: $h = \frac{6}{5}$, $x_0 = \frac{8}{5}$, $y_0^2 = 1 - \frac{16}{25} = \frac{9}{25}$, $r_2^2 = \left(\frac{2}{5}\right)^2 + \frac{9}{25} = \frac{13}{25}$ (the two points $(x_0, \pm y_0)$ give double contact)

Common chord $C_1 - C_2 = 0$: $\left(x + \frac{3}{10}\right)^2 - \left(x - \frac{6}{5}\right)^2 = \frac{97}{100} - \frac{52}{100} \Rightarrow 3x = \frac{9}{5} \Rightarrow x = \frac{3}{5}$

Distance of the chord from the centre of C_2 is $\frac{3}{5}$, so length $= 2\sqrt{\frac{13}{25} - \frac{9}{25}} = \frac{4}{5}$

Final Answer: 0.80